PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 6

One signal, two views

Time series in the time and frequency domains

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[0.5 min · by 0:30] **Title.**

Almost everything we measure in physics is recorded in time: a voltage, a position, a temperature, a blood-flow signal, the output of our own ODE solver. Today we learn to look at such a record in two ways: as values changing in time, and as a sum of oscillations. The second view — the frequency domain — is the main part of the lecture.

Say: "Same numbers, two views. Each shows what the other hides."

Where we left off

From Lectures 4 and 5

- Interpolation flattens fast oscillations and can shift phases
- Padding is extrapolation
- Differentiation amplifies fast fluctuations; integration accumulates slow ones
- Correlated values: uncertainties and p-values too small; shuffling tests independence

Today

- What a time series is, and what we can read from it in time
- The same series as a sum of oscillations: Fourier analysis
- What sampling, record length and windows do to a spectrum
- When the oscillations change in time
- Is a peak in a spectrum real?

Today: what "fast" and "slow" mean, and what padding does to a spectrum.

[1.5 min · by 2:00] **Where we left off.**

Left card: in Lecture 4 we saw that interpolation cuts off the peaks of fast oscillations, that padding a record is a form of extrapolation, and that differentiation amplifies fast fluctuations while integration accumulates slow ones. Each time I said "fast" and "slow" without making it precise — today we do. In Lecture 5: if consecutive values are correlated, uncertainties and p-values are too small; shuffling the order tests whether they are.

Right card: Part A is the time domain — what a time series is, and what we can read directly. Parts B and C are the frequency domain, the main part.

Say: "Today 'fast' and 'slow' get a number: a frequency."

Three views of a time series

View	Question	Tools	Part
Time domain	How does the value change from one moment to the next?	Plot in order; running mean and spread; autocorrelation	A
Frequency domain	Which oscillations is the series made of, and how strong are they?	Discrete Fourier transform (FFT); spectrum	B
Time and frequency	Do these oscillations change in time?	Short-time Fourier transform	C

The views are not competitors

The frequency domain contains the same information as the time series — arranged differently. Some questions are easy in one view and hard in the other.

[1.5 min · by 3:30] **Three views.**

Read the question column. The time domain answers "what happens when"; the frequency domain answers "which rhythms, and how strong"; the third view asks whether the rhythms themselves change.

Callout: the Fourier transform can be inverted exactly, so no information is gained or lost — it is rearranged. What changes is what we can *see*. Two rhythms of similar size, buried in noise, are hard to separate by eye in time and obvious in frequency. A single event — a spike at $t = 12\,\text{s}$ — is obvious in time and spread over all frequencies in the spectrum.

Running example (board): a simulated recording, sampled every 0.1 s for 60 s: two oscillations with periods 4 s and 0.91 s — similar to breathing and heartbeat — plus noise. Simulated, so that we know the truth.

PART A

The time domain

What is a time series, and what can we read from it directly?

[0.5 min · by 4:00] **Part A divider.**

Before any transform: look at the values in time. Everything in Part B assumes things — uniform sampling, stationarity — that we check here.

What is a time series?

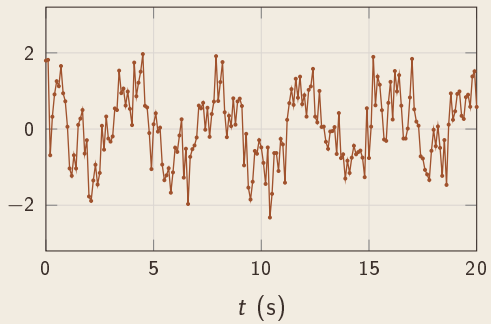
A time series

Values of one quantity at successive times, kept in time order:

$$x_n = x(t_n), \quad t_n = n\Delta t, \quad n = 0, \ldots, N-1$$

- Δt: sampling interval; $f_s = 1/\Delta t$: sampling rate
- N values; duration $T = N\Delta t$
- Today: uniform sampling (irregular: Lecture 4)

Running example: first 20 s of 60 s



Two numbers set the limits of everything that follows: Δt and T.

[2 min · by 6:00] What is a time series?

Definition: the essential word is *order*. A set of 600 values is a sample; the same 600 values with their times are a time series. Examples: a thermometer read every minute, the position of a pendulum every 10 ms, the output of your RK4 solver of Lecture 2 at each step.

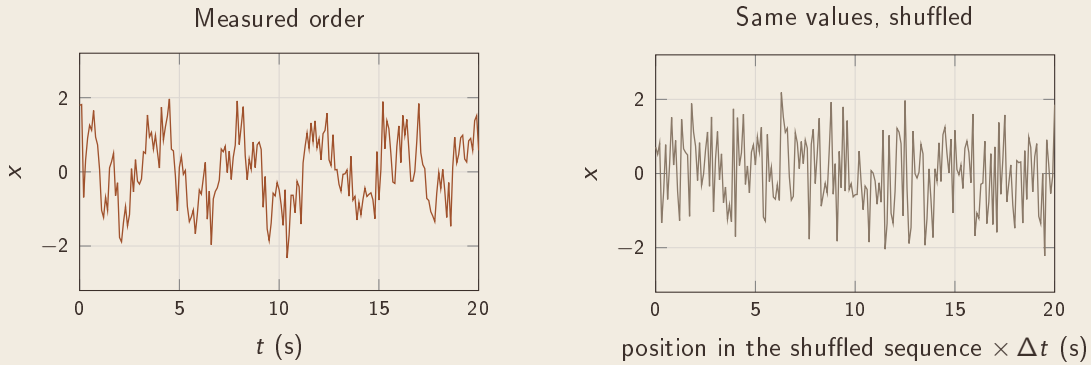
Symbols (point at the list): Δt is the time between samples; its inverse f_s is the number of samples per second. N values cover a duration $T = N\Delta t$. Running example: $\Delta t = 0.1$ s, $f_s = 10$ per second, $N = 600$, $T = 60$ s.

Uniform sampling: today we assume equal spacing. Many records are not uniform — heartbeat times, astronomical observations with gaps. Lecture 4 warned that resampling them changes amplitudes and phases.

Plot: the running example. Can you see the two rhythms of periods 4 s and 0.91 s? The slow one, roughly; the fast one, hardly.

Bottom line — write on the board: Δt limits the fastest change we can see; T limits the slowest. We will make both precise in Part B.

The order carries the information



Same mean, same s, same histogram. Everything that distinguishes the two lies in the order: rhythms, trends, memory. The rest of this lecture is about reading the order.

[1.5 min · by 7:30] **The order carries the information.**

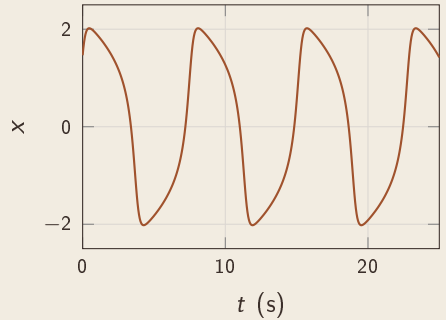
Right panel: the same 600 values, randomly reordered — the shuffle surrogate of Lecture 5. The mean (-0.006), the standard deviation (0.94) and the histogram are identical, because they do not depend on the order.

Point: every description in Lecture 5 — mean, median, s, histogram — is blind to the order. A time series contains more: whether values follow each other smoothly, repeat, drift. That extra information is what time-series analysis is about, and it is what dynamics leaves in data.

Say: "If shuffling would not change your result, you have not analysed a time series — you have analysed a set of numbers."

What we can read directly in time

van der Pol oscillator, $\mu = 2$ (RK4)



$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, sampled every $\Delta t = 0.05\,\mathrm{s}$

Feature	How, and the value here
Amplitude	largest $\lvert x \rvert$: 2.02
Period	between upward zero crossings: 7.630 s (51 periods)
Shape	slow stretches, fast jumps: *not* a cosine
Trend, events	running mean; look for steps, spikes
Memory	autocorrelation (next slides)
Delay	between two signals: cross-correlation

[2 min · by 9:30] What we can read directly in time.

The van der Pol oscillator: a nonlinear, autonomous, deterministic oscillator (Lecture 2 classification), solved with RK4, step $h = 0.001$, and stored every 0.05 — a time series from a simulation. I take time in seconds.

Read the table: amplitude and period are read straight from the series. The period from upward zero crossings (interpolated linearly between the two samples around each crossing — Lecture 4) is 7.630 s, the same for all 51 periods to 4×10^{-4} s: a perfectly regular oscillation.

Shape: the waveform is far from a cosine: slow stretches and fast jumps. Remember this — in Part B it will produce a surprise in the spectrum.

Trends and events: a running mean over a window of several periods shows slow changes; steps and spikes are seen best by eye, in time.

Last two rows: memory and delays — next slides. Cross-correlation is the same idea as autocorrelation, with a second signal: the shift that best aligns the two is their delay.

Stationary or not?

Stationary

A series is **stationary** if its statistical properties — mean, spread, autocorrelation — do not change in time.

Window	Running example mean / s	Slow amplitude grows in time mean / s	Slow drift added mean / s
0–20 s	0.04 / 0.93	0.04 / 0.74	0.29 / 0.95
20–40 s	−0.03 / 0.94	−0.03 / 0.95	0.72 / 0.96
40–60 s	−0.03 / 0.94	−0.03 / 1.17	1.22 / 0.94

- A parameter varying explicitly in time (nonautonomous, Lecture 2) generally makes the series nonstationary
- Judged relative to the record: a variation slower than T looks like a drift

[2 min · by 11:30] Stationary or not?

Definition: not "nothing changes" — the values change all the time — but the *rules* by which they change stay the same: the same mean, spread and correlations in every part of the record.

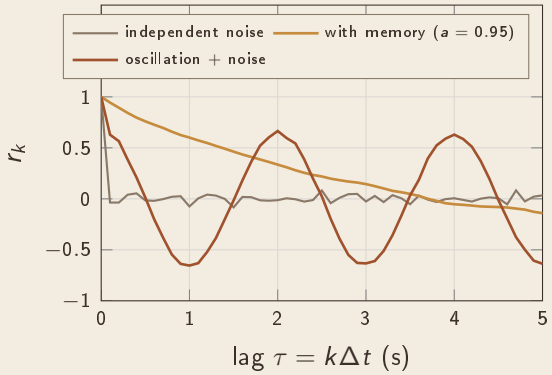
Read the table: cut each 60-s record into three windows of 20 s. Running example: the same mean and spread in every window — consistent with stationarity. Middle column: the amplitude of the slow oscillation grows from 0.4 to 1.6 during the record — a parameter changing explicitly in time. The mean stays at zero, but s grows from 0.74 to 1.17: nonstationary. Right column: a slow drift of 1.5 over the record: the mean changes, s does not.

Why it matters: the spectrum in Part B averages over the whole record. For a nonstationary series it describes no moment in particular. Part C deals with that.

Caution (second bullet): a slow oscillation with a period of 10 minutes, recorded for 60 seconds, is indistinguishable from a drift. Whether to remove a "trend" (detrending) is a decision about what you consider part of the signal: what you remove you can no longer analyse.

Windows: each must contain several periods of the rhythms present. Here they are 20 s, five periods of the slow rhythm; a 2-s window would show a "changing mean" that is just the oscillation.

Autocorrelation: how long does the series remember?



$$r_k = \frac{\sum_{n=0}^{N-1-k}(x_n - \bar{x})(x_{n+k} - \bar{x})}{\sum_{n=0}^{N-1}(x_n - \bar{x})^2}$$

Correlation of the series with itself shifted by k samples.

Memory: $x_n = a\,x_{n-1} + \varepsilon_n$, with independent ε_n. Then $r_k \approx a^k = e^{-\tau/\tau_c}$, correlation time $\tau_c = 1.95\,\mathrm{s}$.

Consequence: 600 values carry the information of about 15 independent ones; the uncertainty of the mean is $6 \times s/\sqrt{N}$.

[2.5 min · by 14:00] **Autocorrelation.**

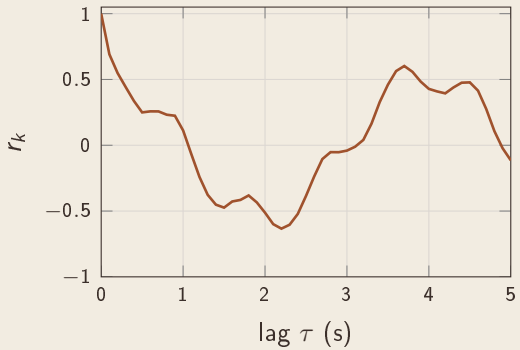
Formula: multiply each deviation from the mean by the deviation k samples later and average; divide by the variance so that $r_0 = 1$. r_k near 1: values $k\Delta t$ apart are nearly the same; near 0: unrelated; negative: opposite.

Three curves, 600 values each, $\Delta t = 0.1\,\mathrm{s}$: *Independent noise* (Lecture 4's definition of noise): $r_k \approx 0$ for every $k \geq 1$; the small wiggles are of size $1/\sqrt{N} = 0.04$. *Memory:* each value is $a = 0.95$ times the previous one plus a new independent random term. The correlation decays as a^k; it falls to $1/e$ after $\tau_c = -\Delta t/\ln a = 1.95\,\mathrm{s}$. A random process, but not independent from sample to sample. *Oscillation of period 2 s plus noise:* r_k oscillates with the same period — a rhythm shows in the autocorrelation even when hard to see in the series. It starts at 0.63, not 1, because the noise is uncorrelated and lowers r_k at every nonzero lag.

Consequence (Lecture 5 made quantitative): for the series with memory, $N_{\mathrm{eff}} \approx N(1-a)/(1+a) = 600 \times 0.05/1.95 = 15$. The standard deviation of the mean is $\sqrt{600/15} \approx 6$ times larger than $s/\sqrt{N}$. Checked: in 4000 simulated series, the means scatter 6.1 times more than $s/\sqrt{N}$.

What the time domain does not separate

Autocorrelation of the running example



The running example holds two rhythms, periods 4 s and 0.91 s, plus noise. In time:

- the slow rhythm dominates the series and its autocorrelation
- the fast one shows only as small ripples
- their sizes cannot be read off

Next

Write the series as a **sum of oscillations**, and measure each one separately.

[1 min · by 15:00] What the time domain does not separate.

The autocorrelation of the running example is dominated by the slow rhythm: minimum near 2 s ($r = -0.51$), maximum near 4 s ($r = 0.43$). The 0.91-s rhythm is visible only as ripples. And neither the series nor its autocorrelation tells us directly that the amplitudes are 1 and 0.5.

Transition: when several rhythms are superposed, we need a description in which each rhythm has its own entry. That is the frequency domain.

PART B

The frequency domain

The same series as a sum of oscillations.

[0.5 min · by 15:30] **Part B divider.**

The main part of the lecture: what the Fourier transform is, how to compute it, and what sampling, record length and windows do to the result.

One oscillation: frequency, amplitude, phase

$$x(t) = A\cos(2\pi f t + \varphi)$$

Symbol	Name	Meaning
f	frequency (Hz $= \mathrm{s}^{-1}$)	oscillations per second; period $P = 1/f$
$\omega = 2\pi f$	angular frequency (rad s^{-1})	phase advance per second
A	amplitude	largest deviation from zero
φ	phase (rad)	where in its cycle the oscillation is at $t = 0$

Fourier's idea, for sampled data

Any N sampled values can be written **exactly** as a sum of cosines at the frequencies $f_k = k/T$, $k = 0, 1, \ldots, N/2$, each with its own amplitude A_k and phase φ_k.

[2 min · by 17:30] **One oscillation.**

Table: four symbols. Frequency in hertz, oscillations per second; period $P = 1/f$. Running example: periods 4 s and 0.91 s are frequencies 0.25 Hz and 1.1 Hz. The angular frequency $\omega = 2\pi f$ is what appears in ODEs ($\ddot{x} = -\omega^2 x$). Phase: two oscillations with the same f and A but different φ are shifted in time by $\varphi/(2\pi f)$.

Fourier's idea (callout): this is exact, not an approximation. N real values are rewritten as $N/2 + 1$ amplitudes and phases — the same number of numbers, N in total (the components at $k = 0$ and $k = N/2$ have no phase of their own). The frequencies are fixed by the record: multiples of $1/T$, the oscillations that fit a whole number of times into the record.

Interpretation, with care: the decomposition always exists. Whether a component corresponds to a physical oscillation is a separate question — we will meet components that do not (harmonics, leakage, noise).

The discrete Fourier transform

$$X_k = \sum_{n=0}^{N-1} x_n \, e^{-2\pi i \, kn/N}, \qquad x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \, e^{2\pi i \, kn/N}$$

- X_k is complex: one number holds amplitude and phase
- Amplitude $A_k = 2|X_k|/N$, phase $\varphi_k = \arg X_k$ (for $0 < k < N/2$)
- Frequencies $f_k = k/T$: spacing $\Delta f = 1/T$
- Largest frequency $f_{N/2} = 1/(2\Delta t)$

FFT: the same numbers, faster

Direct sum: N^2 operations. Fast Fourier transform: about $N \log_2 N$.
For $N = 2^{20}$ (about 10^6): 10^{12} against 2×10^7 — 5×10^4 times fewer. Measured here: 0.05 s.

$e^{-2\pi ikn/N} = \cos(2\pi f_k t_n) - i \sin(2\pi f_k t_n)$: each X_k measures how much of the oscillation at f_k the series contains.

[2.5 min · by 20:00] The discrete Fourier transform.

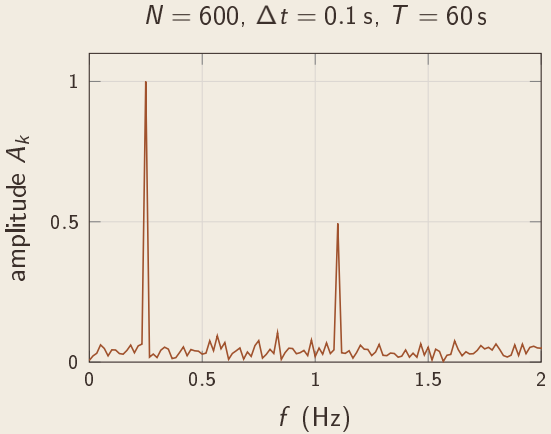
Formulas: X_k is a sum over the data of x_n times a cosine and a sine at frequency f_k — it compares the series with that oscillation. Bottom line of the slide: $kn/N = f_k t_n$, because $f_k = k/(N\Delta t)$ and $t_n = n\Delta t$. The second formula is the inverse: the series rebuilt from the X_k, exactly.

Why the $2/N$: a pure cosine of amplitude A at f_k gives $|X_k| = AN/2$ (half at $+f_k$, half at the mirror frequency $k \to N - k$). So $A_k = 2|X_k|/N$ returns A. For real data the values for $k > N/2$ mirror those below and carry nothing new.

Two limits, both set by the record: spacing $\Delta f = 1/T$; largest frequency $1/(2\Delta t)$, half the sampling rate — slide after next. Running example: $\Delta f = 1/60$ Hz, largest 5 Hz.

FFT (callout): the fast Fourier transform (Cooley and Tukey, 1965; Gauss had the idea in 1805) gives *exactly* the same X_k, reusing partial sums. Complexity as in Lecture 1: N^2 against $N \log_2 N$. A million points: 10^{12} operations would take about 20 minutes at 10^9 per second; the FFT on this laptop took 0.05 s. In Python: `numpy.fft.rfft(x)` for real data.

The spectrum of the running example



	true	from X_k
A at 0.25 Hz	1	1.000
A at 1.1 Hz	0.5	0.494
φ at 1.1 Hz	1.00	1.06
noise level	—	≈ 0.03

Parseval: the variance splits over frequencies. Here 0.874 in total: 0.622 in the two peaks, 0.252 spread by the noise.

Two normalisations: amplitude A_k for peaks; **power spectral density** $S_k = 2\Delta t|X_k|^2/N$ (units x^2/Hz) for broad spectra — its area is the variance.

[2.5 min · by 22:30] **The spectrum of the running example.**

What was hard in time is obvious here: two sharp peaks, at 0.25 and 1.1 Hz, with heights 1.000 and 0.494 — the amplitudes we put in. The phase of the 1.1-Hz component comes out as 1.06 rad against the true 1.00. Nothing above 2 Hz but noise (the plot stops there; the spectrum continues to 5 Hz).

Noise: independent noise (Lecture 4's definition) has no preferred frequency; it spreads evenly over all 300 frequencies, each with a small amplitude, about 0.03 here. A rhythm concentrates its variance at one frequency; noise spreads it.

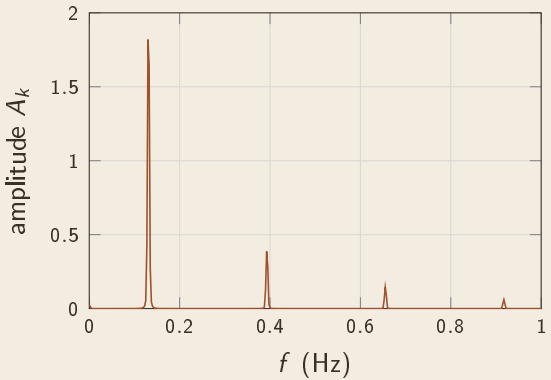
Parseval: the sum of the squared amplitudes over two equals the variance — the spectrum is a breakdown of the variance by frequency. Total 0.874 = the variance of the data; $1^2/2 + 0.5^2/2 = 0.625$ expected in the peaks, 0.622 found; the rest, 0.252, is the noise ($\sigma^2 = 0.25$).

Normalisations — a practical warning: software uses different conventions (amplitude, power, power per hertz, one- or two-sided). Before quoting a number, check it with Parseval: does the sum (or the area) give the variance? For a sharp peak use the amplitude; for broad features — noise, Part C — use the density, whose value does not depend on T.

Peaks exactly on the grid: 0.25 = 15/60 and 1.1 = 66/60, so each falls exactly on an f_k. In general they do not — slide on leakage.

A peak is not always a separate oscillation

van der Pol, 400 s record (Hann window)



f (Hz)	multiple of f_0	amplitude
0.131	1	1.82
0.393	3	0.39
0.655	5	0.14
0.917	7	0.06

$f_0 = 1/7.630\,\text{s} = 0.131\,\text{Hz}$. Even multiples: zero, because $x(t + P/2) = -x(t)$.

One oscillation, many peaks: the **harmonics** describe its non-sinusoidal shape.

[1 min · by 23:30] **A peak is not always a separate oscillation.**

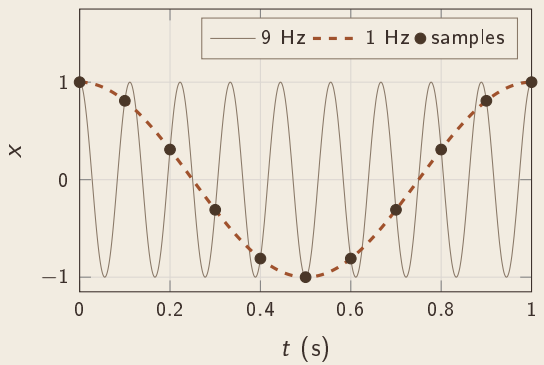
The van der Pol series of slide 7 — one oscillator, one period, 7.630 s. Its spectrum has peaks at f_0, $3f_0$, $5f_0$, $7f_0$. These are the cosines needed to build the slow-stretch, fast-jump shape. A pure cosine has only f_0.

Even harmonics vanish: the second half of each period is the first half upside down; such a waveform contains only odd multiples. Measured: below 10^{-3}.

Window: here I used a Hann window — explained on slide 18 — because 400 s is not a whole number of periods.

Message: peaks at exact multiples of one frequency usually mean one nonlinear oscillation, not several oscillators. Check in time: one period, a non-sinusoidal shape.

Sampling: the fastest oscillation Δt can see



Sampling at $f_s = 10$ per second: the samples of a 9-Hz cosine lie exactly on a 1-Hz cosine.

Nyquist frequency

$f_N = \dfrac{f_s}{2} = \dfrac{1}{2\Delta t}$. A component above f_N appears at the wrong frequency $|f - mf_s|$, for the integer m that brings it below f_N: **aliasing**.

Cannot be undone after sampling: sample faster, or filter out fast components *before* sampling.

[2 min · by 25:30] Sampling and aliasing.

Plot: the grey curve oscillates 9 times per second. Sampled every 0.1 s, the dots fall exactly on the dashed curve, which oscillates once per second. From the samples alone the two are indistinguishable: $\cos(2\pi \cdot 9 \cdot n/10) = \cos(2\pi n - 2\pi n/10) = \cos(2\pi n/10)$.

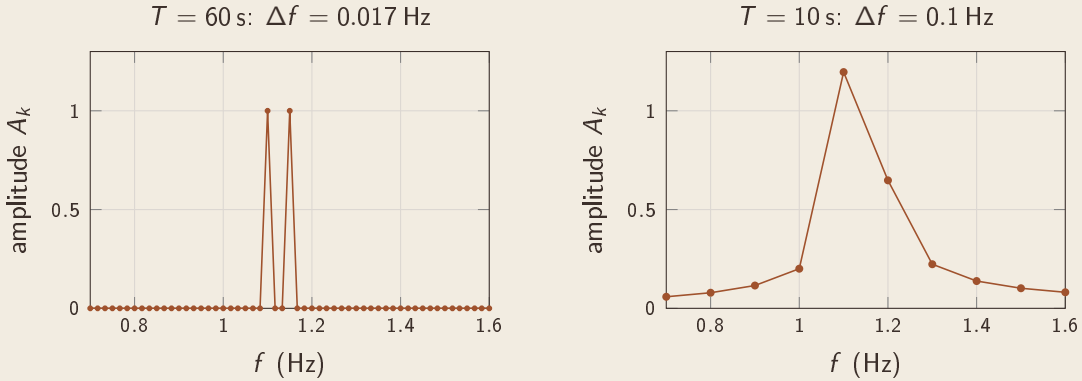
Rule: with Δt we can identify frequencies up to $f_N = 1/(2\Delta t)$ — two samples per period. Faster components are not lost; worse, they reappear as slower ones: 9 Hz at 1 Hz, 11 Hz at 1 Hz, 6 Hz at 4 Hz.

Familiar examples: wagon wheels turning backwards in films (24 frames per second); a strobe light freezing a fan.

Why it cannot be undone: after sampling, the 1-Hz content of the samples is a mixture of real 1-Hz content and aliased 9-Hz, 11-Hz, … content. Instruments therefore filter out fast components electronically before sampling (an anti-aliasing filter).

Two numbers to keep apart: two samples per period are enough to *identify* a frequency; about ten are needed to *reproduce* the shape by interpolation (Lecture 4). In a simulation the same applies to the output interval: storing every 100th step of an RK4 solution is sampling.

Record length: how close two frequencies can be



$\cos(2\pi\,1.10\,t) + \cos(2\pi\,1.15\,t)$. **Resolution** $\Delta f = 1/T$: to separate frequencies δf apart, record for $T \gtrsim 1/\delta f$ — here 20 s. The lowest frequency we can see is also about $1/T$.

[1.5 min · by 27:00] **Record length.**

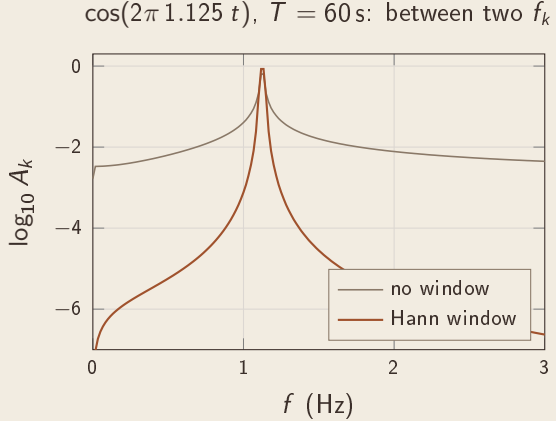
Two oscillations, 0.05 Hz apart, amplitude 1 each. Left, 60 s of data: spectrum values every 1/60 Hz; two clean peaks, both of height 1.00. Right, 10 s of data: values only every 0.1 Hz; one lump around 1.1 Hz. Nothing in the analysis can split it.

Why (time picture): the two oscillations drift apart by one full cycle every $1/\delta f = 20$ s — they beat. In 10 s they hardly drift apart, so they look like one oscillation with a slowly changing amplitude.

Rule: Δt sets the highest frequency, T the finest spacing and the lowest frequency. A rhythm with a period of 10 minutes needs a record of several 10 minutes — otherwise, as in Part A, it is a "drift".

Question: "How long must you record to separate 1.10 Hz from 1.15 Hz?" At least 20 s; in practice several times longer, because noise and windows broaden peaks.

Leakage: the record does not hold whole periods



The DFT treats the record as one period of a periodic signal. A frequency off the grid makes a jump where the end meets the start, and its power **leaks** to all frequencies.

Window: multiply by $w_n = \sin^2(\pi n/N)$ (Hann), which goes smoothly to zero at both ends.

	none	Hann
peak (true 1)	0.64	0.85
0.5 Hz away	1.3×10^{-2}	1.3×10^{-5}

Cost: each peak about two Δf wide.

[2 min · by 29:00] **Leakage.**

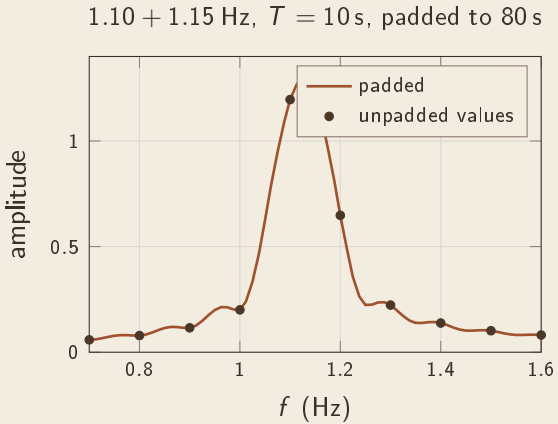
Why: the DFT uses only frequencies that fit a whole number of times into T; implicitly it treats the record as repeating forever. A cosine at 1.125 Hz completes 67.5 cycles in 60 s; repeated, the end joins the start with a jump. That jump is not in the physics; it is the periodic padding of Lecture 4, the worst of the paddings. Representing a jump needs all frequencies, so the peak leaks.

Plot (log scale): without a window, the leakage falls off slowly: at 0.5 Hz from the peak it is still 1.3×10^{-2} — a real oscillation of amplitude 0.01 there would be hidden. The peak itself is 36% too low, because the true frequency lies between two grid points.

Hann window: multiply the data by a smooth bump that is zero at both ends; the jump disappears. Leakage at 0.5 Hz: 1.3×10^{-5}, a thousand times lower; the peak error falls to 15%. The amplitude is corrected by dividing by $\sum w_n$ instead of N.

Cost: the window reduces the effective length of the record, so a peak exactly on the grid now occupies three values (0.5, 1, 0.5) instead of one: resolution is slightly worse. **Rule**: use a window by default; skip it only when the frequencies are known to fit the record exactly, as in the running example.

Zero padding: a finer grid, not a finer resolution



Append zeros to the record: $N \to 8N$.

- The grid becomes 8 times finer: the spectrum is **interpolated** between the original values
- Still one peak: resolution stays $1/T$, set by the data
- Zeros are an assumption about the signal outside the record (Lecture 4: padding is extrapolation)

Uses: a smoother plot, a better located peak, a convenient length for the FFT.

[1.5 min · by 30:30] **Zero padding.**

The promise of Lecture 4. Take the 10-s record that could not separate 1.10 from 1.15 Hz, append 70 s of zeros and transform 80 s. The spectrum now has values every 0.0125 Hz — the smooth curve — and it passes exactly through the original values (dots). One peak, at 1.125 Hz. Compare with the real 60-s record of the previous slide: two peaks.

Why: the zeros add no information about the signal. Padding computes the same spectrum at intermediate frequencies: it is interpolation in frequency.

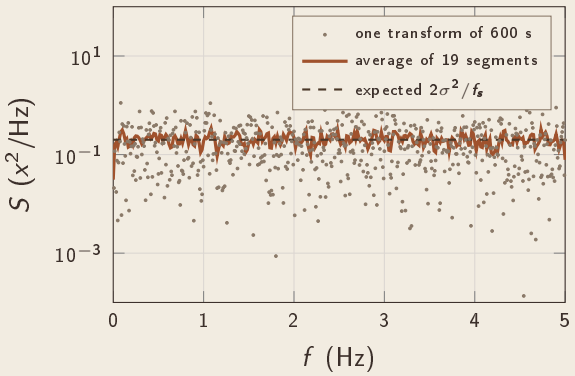
Lecture 4 connection: padding with zeros assumes the signal is zero outside the record — the rectangular "no window" case, with its jumps to zero at both ends. Window first, then pad.

Legitimate uses: a smoother curve to read a peak position more precisely (here the true 1.125 is the centre of the lump); a length that is a power of two (modern FFT routines handle any N, so this matters less than it used to).

Say: "Padding makes the picture smoother, not sharper."

Noise in the spectrum: more data do not smooth it

Independent noise, $\sigma = 1$, $T = 600\,\mathrm{s}$



Estimate	scatter	Δf (Hz)
one transform, 60 s	93%	0.017
one transform, 600 s	102%	0.0017
19 segments of 60 s	25%	0.017
119 segments of 10 s	8%	0.1

A longer record gives *more* values, each as uncertain. **Welch's method**: cut into windowed, overlapping segments and average their spectra. Scatter falls as $1/\sqrt{\text{number of segments}}$; resolution becomes $1/T_{\text{segment}}$.

[2 min · by 32:30] **Noise in the spectrum.**

Independent noise has a flat spectrum (it is often called *white*): the same expected density at all frequencies, $2\sigma^2/f_s = 0.2$ here, so that the area from 0 to 5 Hz is $\sigma^2 = 1$. This is the frequency-domain version of Lecture 4's definition.

The surprise (grey dots, every fifth value shown): the single transform scatters by about 100% around the expected level — some values ten times below it. With 60 s: 93%; with 600 s: 102%. Ten times more data, no improvement. Each X_k is a sum of N terms, so the data average down; but the number of X_k grows with N too, and each still estimates one random amplitude. (For Gaussian noise each value follows an exponential distribution, whose spread equals its mean.)

Welch's method (red line): cut the 600 s into 60-s segments overlapping by half, apply a Hann window to each, transform, and average the 19 spectra. Scatter 25%. Shorter segments, 10 s: 119 segments, 8% scatter, but resolution 0.1 Hz. The Lecture 5 trade-off: average independent estimates to reduce the uncertainty — paid for with resolution.

Rule: for a broad spectrum, average segments; for a sharp peak, a long single transform. Python: `scipy.signal.welch`.

Autocorrelation and spectrum: two views of the same information

Series	Autocorrelation r_k	Power spectral density S
Independent noise	0 for every $k \geq 1$	flat
With memory, τ_c	decays over τ_c	largest at low f, falls above $f \approx 1/(2\pi\tau_c)$
Oscillation at f_0	oscillates with period $1/f_0$	peak at f_0
Oscillation that loses its phase	oscillation decays	peak at f_0, broadened

Wiener–Khinchin theorem

For a stationary series, the power spectral density is the Fourier transform of the autocorrelation. Neither contains the phases: different series can share both.

[1.5 min · by 34:00] **Autocorrelation and spectrum.**

Read the table row by row, pointing back at the curves of slide 9. Independent noise: no memory, no preferred frequency. A series with memory, like the $a = 0.95$ example, changes slowly: its variance is at low frequencies, with a corner near $1/(2\pi\tau_c) = 0.08\,\text{Hz}$ (often called *red* noise — more power at low frequency). An oscillation: periodic autocorrelation, a peak. An oscillation that gradually forgets its phase: the autocorrelation oscillates but decays, and the peak is broad — the faster it forgets, the broader. Remember this row for Part C.

Theorem: the two are a Fourier pair — they contain the same information. Choose the view in which the question is easier.

What both lose: the phases φ_k. Keep all amplitudes, choose new random phases, transform back: a different series with the same spectrum and autocorrelation. Such "Fourier surrogates" are used to test for properties that depend on the phases, for example nonlinearity.

Differentiation, integration and averaging are filters

Each operation multiplies the component at frequency f by a factor $H(f)$:

Operation ($\Delta t = 0.1$ s)	$H(f)$	0.25 Hz	1.1 Hz	5 Hz
Exact derivative	$2\pi f$	1.57	6.91	31.4
Central difference	$\sin(2\pi f \Delta t)/\Delta t$	1.56	6.37	0
Exact integral	$1/(2\pi f)$	0.64	0.14	0.03
Average of 10 samples (1 s)	$\dfrac{\sin(10\pi f \Delta t)}{10\sin(\pi f \Delta t)}$	0.90	−0.09	0

- Lecture 4, made precise: the derivative **amplifies high frequencies** — noise; the integral amplifies low ones; an offset is $f = 0$
- The moving average passes slow components, removes fast ones — and flips the sign of the 1.1-Hz one
- Any filter: multiply X_k by $H(f_k)$, transform back. Phases change too

[2 min · by 36:00] Operations as filters.

Why a factor: differentiate $A\cos(2\pi ft)$ and get $2\pi fA$ times a sine — the same frequency, amplitude multiplied by $2\pi f$, phase shifted by 90°. Integration divides by $2\pi f$. Linear operations act on each frequency separately; that is why the frequency domain is natural for them.

Read the rows: exact derivative: the 1.1-Hz component grows 6.9 times, noise at 5 Hz 31 times — Lecture 4's amplified noise. The central difference of Lecture 4 is accurate at low f (1.56 against 1.57), 8% low at 1.1 Hz (9 samples per period), and zero at the Nyquist frequency: neighbours two steps apart are equal there. Integral: the slow component wins; at $f = 0$, an offset, $1/(2\pi f)$ is infinite — the offset that grew as bT in Lecture 4.

Moving average (last row): 10 samples, 1 s. The 0.25-Hz component passes at 90%, 5 Hz is removed. But 1.1 Hz comes out at −0.09: 9% of the amplitude, *inverted*. Checked on the data: amplitude $0.5 \to 0.046$, in antiphase. A "smoothing" that inverts a rhythm. The interpolation of Lecture 4 is also a filter — it was flattening fast oscillations.

Last bullet: filtering in the frequency domain is multiplication. Filters generally shift phases; compare signals only after the same filter.

PART C

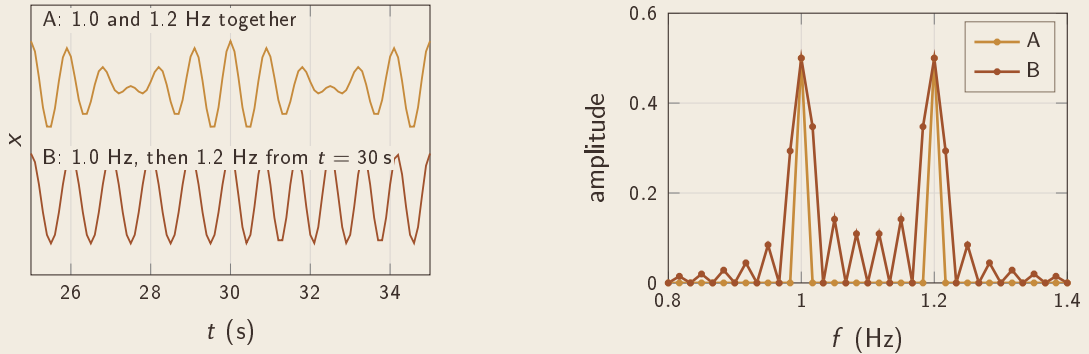
When the oscillations change in time

A spectrum averages over the whole record. What if the rhythms do not stay the same?

[0.5 min · by 36:30] **Part C divider.**

Everything in Part B assumed stationarity: the same rhythms throughout the record. Many physical and biological oscillators are nonautonomous — their frequencies and amplitudes change in time. What does a spectrum show then?

The spectrum says which frequencies, not when



Both: peaks of 0.50 at 1.0 and 1.2 Hz ($T = 60$ s). Two oscillators, or one that changed its frequency? The amplitude spectrum alone cannot say; B's peaks are only wider, because each frequency lasted 30 s.

[1.5 min · by 38:00] **Which frequencies, not when.**

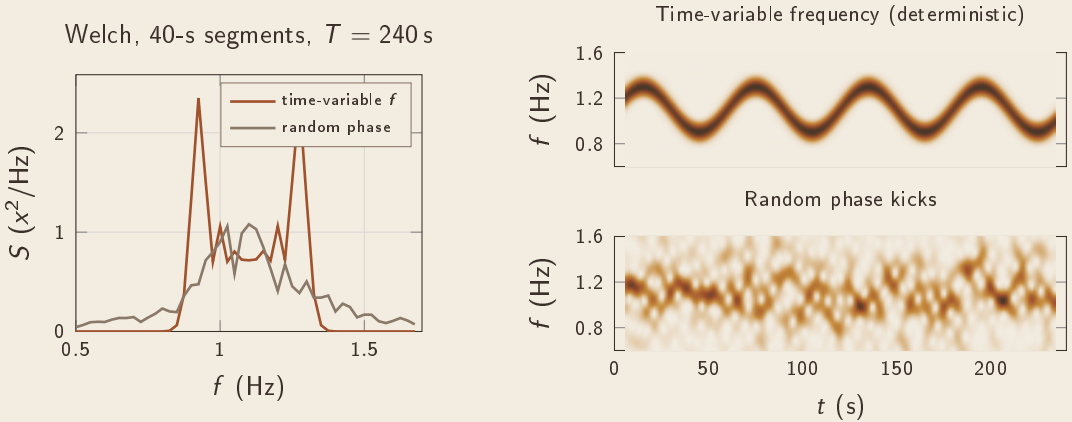
A: two oscillations of amplitude 0.5, present all the time — they beat. B: one oscillation of amplitude 1, which switches from 1.0 to 1.2 Hz halfway through. Physically very different: two systems, or one system whose parameter changed.

Spectra: both have peaks of height 0.50 at the same two frequencies. The only visible difference: B's peaks are broader, because each frequency is present for 30 s, so its resolution is 1/30 s rather than 1/60 s. With noise, that difference would be hard to see.

Where is the information? In the phases — the transform is invertible. But the phases are hard to read. The direct way: look at the frequencies in time — two slides on.

Say: "A spectrum is a summary of the whole record. It answers 'which', not 'when'."

A broad peak is not necessarily noise



Top: $f(t) = 1.1 + 0.2\sin(2\pi t/60\,\text{s})$ Hz, no noise. Bottom: 1.1 Hz, phase perturbed by independent random kicks. Both peaks are broad; only the time–frequency view shows why.

[2 min · by 40:00] **A broad peak is not necessarily noise.**

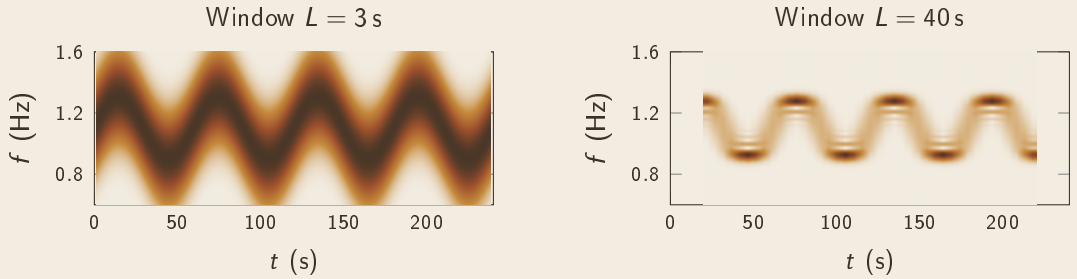
Two oscillators, 240 s each, amplitude 1. Red: deterministic, no noise at all; its frequency is varied in time, between 0.9 and 1.3 Hz with a period of 60 s — a nonautonomous oscillator (Lecture 2), like a heart whose rate is modulated by breathing, seen on its own. Grey: frequency 1.1 Hz, but the phase receives small independent random kicks, so it slowly loses its rhythm (the last row of slide 21).

Left: both spectra are broad: the deterministic one has 96% of its power between 0.9 and 1.3 Hz, the random one 60%. If we read a broad peak as "an oscillation plus noise", we are right for one of them and wrong for the other.

Right — the time–frequency view (next slide explains how): the spectrum computed in a 12-s window that slides along the record. Top: a clean line following the imposed frequency — one oscillation whose frequency changes regularly in time. Bottom: an irregular, fragmented band — the frequency wanders without a pattern.

Message — the theme of Lectures 2 and 3: unexplained width is not automatically noise. Time variability spreads a peak just as well. Look at the frequency in time before deciding.

Short-time Fourier transform: the time–frequency trade-off



Short-time Fourier transform: the windowed DFT of the data within a window of length L, slid along the record. Resolution: about L in time and $1/L$ in frequency — one cannot have both.

L	$1/L$	What we see of $f(t)$ (period 60 s, range 0.9–1.3 Hz)
3 s	0.33 Hz	follows the changes, but blurred in frequency
12 s	0.08 Hz	both (previous slide)
40 s	0.025 Hz	sharp in frequency, but the changes are smeared

[2 min · by 42:00] **Short-time Fourier transform.**

How: take L seconds of data, apply a Hann window, transform, and plot the result as one column at the window's centre time. Slide the window and repeat. The picture is often called a spectrogram. Python: `scipy.signal.stft`.

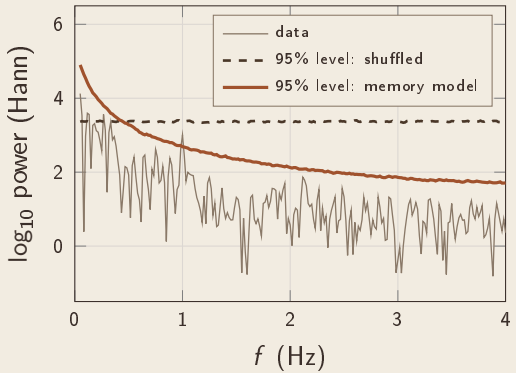
The trade-off: the window has length L, so we cannot say *when* within L something happened; and a record of length L has resolution $1/L$ (slide 17). Shorter windows: sharper in time, blurred in frequency. Left: 3 s — the band is 0.3 Hz wide, nearly as wide as the whole excursion. Right: 40 s — each window contains two-thirds of a modulation cycle, so it sees a mixture of frequencies; the power piles up where the frequency lingers, at the extremes 0.9 and 1.3 Hz.

Choosing L: long enough to resolve the frequencies of interest, short enough that they hardly change within L. With 12 s we had both, because the frequency changes slowly compared with its own period.

Beyond today: when the frequencies of interest span a wide range, one window length cannot suit them all. The *wavelet transform* adapts the window to the period — long for slow oscillations, short for fast ones.

Is a peak real? Choose the null hypothesis carefully

Memory ($a = 0.95$) + 0.15 at 1 Hz, $T = 60$ s



Test the largest peak against surrogates (Lecture 5) of two kinds:

- **shuffled**: H_0 = independent values
- **memory model**: $H_0 = x_n = a\,x_{n-1} + \varepsilon_n$, with a fitted to the data

500 records	shuffled	memory
no oscillation: "real"	100%	3.6%
with oscillation: "real"	100%	49%

Against the wrong H_0, every record has a "real" peak: slow variation is not independent noise.

[2.5 min · by 44:30] Is a peak real?

The data (grey): a random process with memory, $a = 0.95$, as on slide 9, plus a weak oscillation of amplitude 0.15 at 1 Hz; $T = 60$ s. Its spectrum rises steeply towards low frequencies, with a small bump at 1 Hz.

The test: the statistic is the largest ratio, over 0.05–4 Hz, of the data spectrum to the average surrogate spectrum — one number for the whole spectrum, so that we do not run 238 tests, one per frequency (Lecture 5: many tests). The lines show the 95% level of that statistic, from 999 surrogates.

Shuffled surrogates (dashed): their spectrum is flat. The low frequencies of the data exceed the 95% level by up to a factor of 6; $p = 0.001$: "significant" — but for the slow variation; the 1-Hz bump itself stays below this level. Over 500 records *without* any oscillation: 100% false alarms. The test answers a question we did not ask: "are the values independent?" — no.

Memory-model surrogates (red): random series with the same a and variance as the data, a estimated from r_1. The 95% level follows the shape of the spectrum; only the 1-Hz bump exceeds it. False alarms 3.6%, close to the nominal 5%; the weak oscillation is found in 49% of records — an honest power for amplitude 0.15 against a background of standard deviation 1.

Caution: the memory model is a parametric H_0 — check that the autocorrelation of the data looks like a^k. Fourier surrogates (slide 21) keep the whole spectrum, peak included, so they cannot test peaks.

Say: "The p-value is only as meaningful as the H_0 behind it."

Summary

View	Tools	Check
Time	plot in order; amplitude, period, shape; running mean and spread; autocorrelation	stationarity; memory ($N_{\text{eff}} \ll N$); events and drifts
Frequency	DFT via FFT; amplitude spectrum for peaks; Welch density for broad spectra	aliasing ($f < 1/(2\Delta t)$); resolution $1/T$; leakage (window); padding is interpolation; harmonics
Time and frequency	short-time Fourier transform	window L: L in time against $1/L$ in frequency
Significance	surrogates	H_0 that matches the background

[1.5 min · by 46:00] **Summary.**

The slide to photograph. Read the check column: each entry is a way to get a clean-looking but wrong answer — an aliased frequency, two peaks that are one, a leakage tail mistaken for a weak rhythm, a smooth padded peak mistaken for resolution, a harmonic mistaken for a second oscillator, a broad peak mistaken for noise, a "significant" peak tested against the wrong H_0.

Before you compute a spectrum: six questions

1. Is Δt short enough for the fastest component? *(aliasing: was it filtered before sampling?)*
2. Is T long enough for the slowest rhythm and the resolution I need? *($\Delta f = 1/T$)*
3. Is the series stationary, or do its rhythms change in time? *(time–frequency view)*
4. Did I use a window, and is padding showing more than the data contain? *(leakage)*
5. Are the peaks separate oscillations, harmonics, or leakage? *(check in time)*
6. Against which H_0 is a peak significant? *(surrogates that match the background)*

Report Δt, T, window, segments and normalisation — and look again in time.

[1 min · by 47:00] **Six questions.**

Same format as the previous lectures. Questions 1 and 2 are decided before the measurement — afterwards it is too late. Questions 3 to 5 are about reading the spectrum; question 6 about testing it.

Box: a spectrum without Δt, T, window and normalisation cannot be reproduced or compared.

TO TAKE AWAY

A spectrum shows which rhythms, not when.

Δt sets the fastest, T the finest and slowest. Window, then look again in time: rhythms that change are not noise.

[0.5 min · by 47:30] **Take away.**

Return to the running example: in time, a noisy curve; in frequency, two rhythms at 0.25 and 1.1 Hz with amplitudes 1.000 and 0.494, read off directly. That is the power of the frequency domain. Its limits: it averages over the record, it needs Δt and T chosen for the rhythms we seek, and a broad peak may be a rhythm that changes in time.

Exercise pointer: simulate the van der Pol oscillator with RK4, compute its spectrum with and without a window, then make μ change slowly in time and look at the short-time Fourier transform.

Timing buffer: the plan finishes at about 47.5 minutes, leaving 2.5 minutes for questions. If short of time, shorten slides 15, 21 and 25.